DAR ES SALAAM INSTITUTE OF TECHNOLOGY

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MODULE NAME: FUNDAMENTALS OF DATA STRUCTURES AND ALGORITHM FOR TECHNICIANS:

MODULE CODE: COT 05213.

1. Rank the following complexities interm of their rate of the growth starting with the slowest to fastest:

$O(n)$, $O(\log n)$, $O(n^5)$, $O(n \log n)$, $O(1)$.

Answer:

$$O(1) < O(\log n) < O(n) < O(n \log n) < O(n^5).$$

→ Since: to rank these we look at how the functions grow as "n" approaches infinity.

2. Find the Big-O complexity for the following functions:

(a) $4n^2 + 20n + 3$

(b) $25n^2 + 15$

(c) $n^3 + 5\log n + 6$

(d) $10^5 + 10^5 n + 10^5 n^2$

(e) $72n(n + 3n^2)$

(f) $2^8 + 32^4 n - 5$

Answers

Functions	Big-O complexity.
(a) $4n^2 + 20n + 3$	$O(n^2)$
(b) $25n^3 + 15$	$O(n^3)$
(c) $n^3 + 5\log n + 6$	$O(n^3)$
(d) $10^5 + 10^5 n + 10^5 n^2$	$O(n^2)$
(e) $72n(n + 3n^2)$	$O(n^3)$
(f) $2^8 + 32^4 n - 5$	$O(n)$.

3. Design an algorithm that runs in worst-case O(n) time to add all the numbers from 1 to a given number n (including n).

Answer:

→ In order to achieve $\Theta(n)$ complexity, I have to iterate through each number from 1 to n and to add it to a running total.

```
sum := 0
for i := 1 to n do:
    sum := sum + i
endfor
return sum.
```

4. ~~Big-O~~ For each of the following code fragments, give the worst-case running time complexity in Big-O notation.

(a) For this case, give the complexity in term of the variable number.

```
x = 4
for i := 0 to number do:
  x := x + 1
endfor.
```

Answer:

Complexity in terms of number, is O(number).

(b) In the following, assume that the function get_choice() runs in O(1) time and give the complexity in term of the cycles variable.

```
choice := get_choice()
if choice = 1 then:
    print "Ready"
else if choice = 2 then:
  for i := 0 to cycles do:
      print i
  endfor
  endif
endif
```

Answer:

Complexity in terms of cycles is O(cycles).

4 (c) There are two functions below. Find the complexity of the calculate function. (2)

```
increase (val : interger, iter: interger): interger
begin:
    for i:=1 to iter do:
        val : =val +5
    endfor
    return val
end
calculate (val : interger, iter : interger): interger
begin:
    for i:=1 to iter do:
        val : = val + increase (val, iter)
    endfor
    return val
end
```

Answer:

⇒ Complexity : $O(N^2)$ where $N = iter$.

This is a standard nested loop structure where an outer loop of N iterations contains an inner function call that also performs N iterations. The total operations are $N \times N = N^2$

4. (d)

These two methods are almost the same as in question (c) except that in calculate the first line of the for loop is different. Find the complexity of the calculate method.

```
increase (val: integer, iter: integer): integer
begin:
    for i:=1 to iter do:
        val := val + 5
    endfor
    return val
end
calculate (val: integer, iter: integer): integer
begin:
    for i:=1 to iter do:
    val := val + increases (val, i)
    endfor
    return val
end
```

Answer:

Complexity = $O(N^2)$ where N = iter

The work I perform is a summation of $1 + 2 + \ldots N$, which equals $\frac{N(N+1)}{2}$. By the rules of Big O notation I ignore the constant $\frac{1}{2}$ and the lower order term (N), leaving $O(N^2)$ as the dominant growth rate.

5 Fill in the blanks (marked with __) with either $\leq$, $\geq$, or $=$:

(i) For some functions $f(n)$ and $g(n)$, if $f(n)$ is $O(g(n))$, then for some constants c and K, for all $n \geq K$, $f(n)$ $\underline{\;\leq\;}$ $c\,g(n)$.

(ii) For some functions $f(n)$ and $g(n)$, if $f(n)$ is $\Omega(g(n))$, then for some constants c and K, for all $n \geq K$, $f(n)$ $\underline{\;\geq\;}$ $c\,g(n)$.

6. Show that $4n + 12$ is $O(n)$, Hint: Find two suitable constants for c and K

Answer:

$4n + 12 \leq c \cdot n$

let $c = 16$

$K = 1$

$4n + 12 \leq 4n + 12n = 16n.$

This holds true for all $n \geq 1$.

Show that $5n^2 - 16$ is $O(n^2)$. Hint: Find two suitable constants for c and K. Try letting $c = 1$ and solve the inequality for that.

Answer:

$5n^2 - 16 \leq c \cdot n^2$

let

$c = 5$

$K = 1$

$5n^2 - 16 \leq 5n^2$

since $5n^2 - 16$ is always less than $5n^2$, this holds true for all $n \geq 1$.